NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2025 - 2026

Batch Number	AG2
Team Members	K. Thirupatamma(22471A0526) B. Velagini Rani(22471A0504) Y. Mythri(22471A0569)
Guide	Dr. S.V.N. Sreenivasu
Title	Road Traffic Accident Risk Prediction and Key Factor Identification Framework Based on Explainable Deep Learning
Domain/Technology	DEEP LEARNING
Base Paper Link	**https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10658644**
Dataset Link	https://www.kaggle.com/datasets/tsiaras/uk-road-safety-accidents-and-vehicles https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python (COLAB)
Hardware Requirements	SystemType: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	This study introduces a novel deep learning framework for predicting road traffic accident risks and identifying key contributing factors. The proposed CNN–BiLSTM–Attention model captures spatial and temporal accident patterns using a spatial-temporal attention mechanism. Evaluated on real-world UK and US traffic datasets, the model achieves high prediction accuracy with Mean Absolute Errors (MAE) of 0.2475 and 0.2683, respectively. To interpret model predictions, DeepSHAP is employed to rank feature importance, revealing the top 15 influential factors such as speed limit and vehicle count. Reintegrating these key features into the model preserves accuracy while significantly reducing training time. This framework supports improved risk assessment and targeted decision-making for traffic safety management.

Signature of the student(s) **Signature of the Guide** **Signature of the project coordinator**